

Neuroscience of motor control

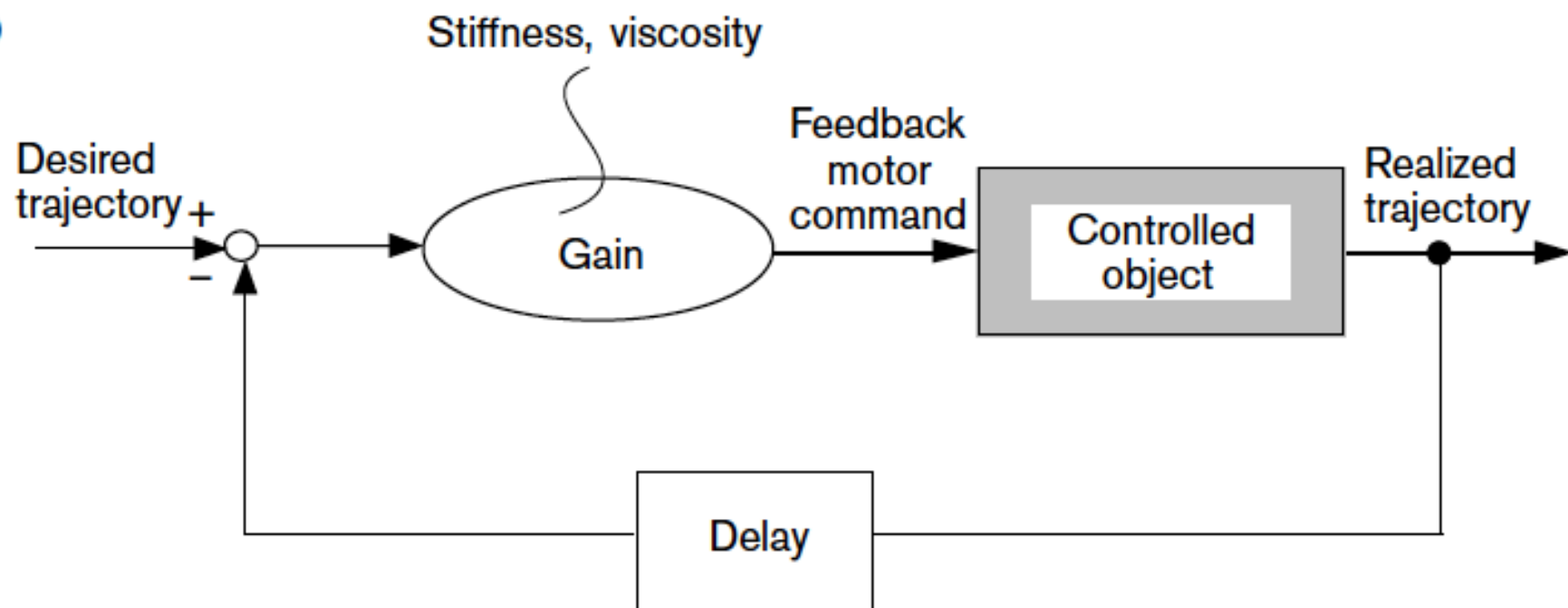
- Nikolai Bernstein – degrees of freedom problem
- Argued against simplistic reflex models
- Was both an experimentalist and a theoretician

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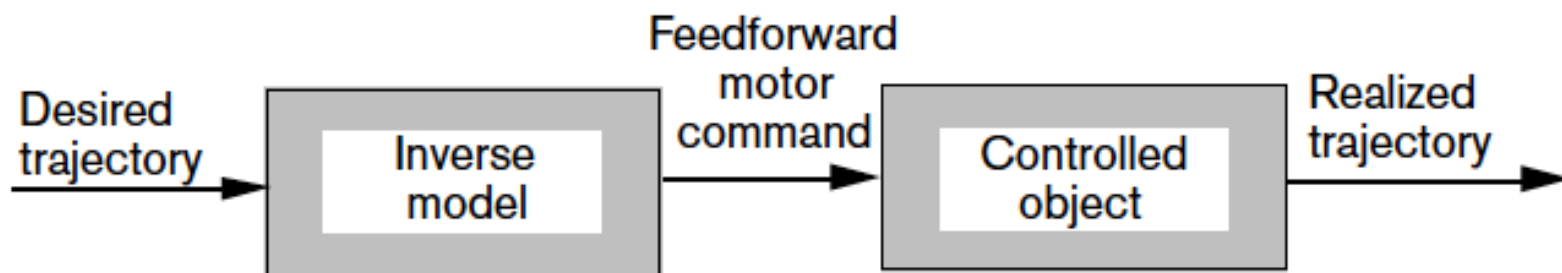
- Internal model hypothesis originated in the control theory literature and proposed in biological context by Ito (1970), Kawato et al (1987), Jordan & Rumelhart (1992) and others.
- Just using feedback control would be too slow to work

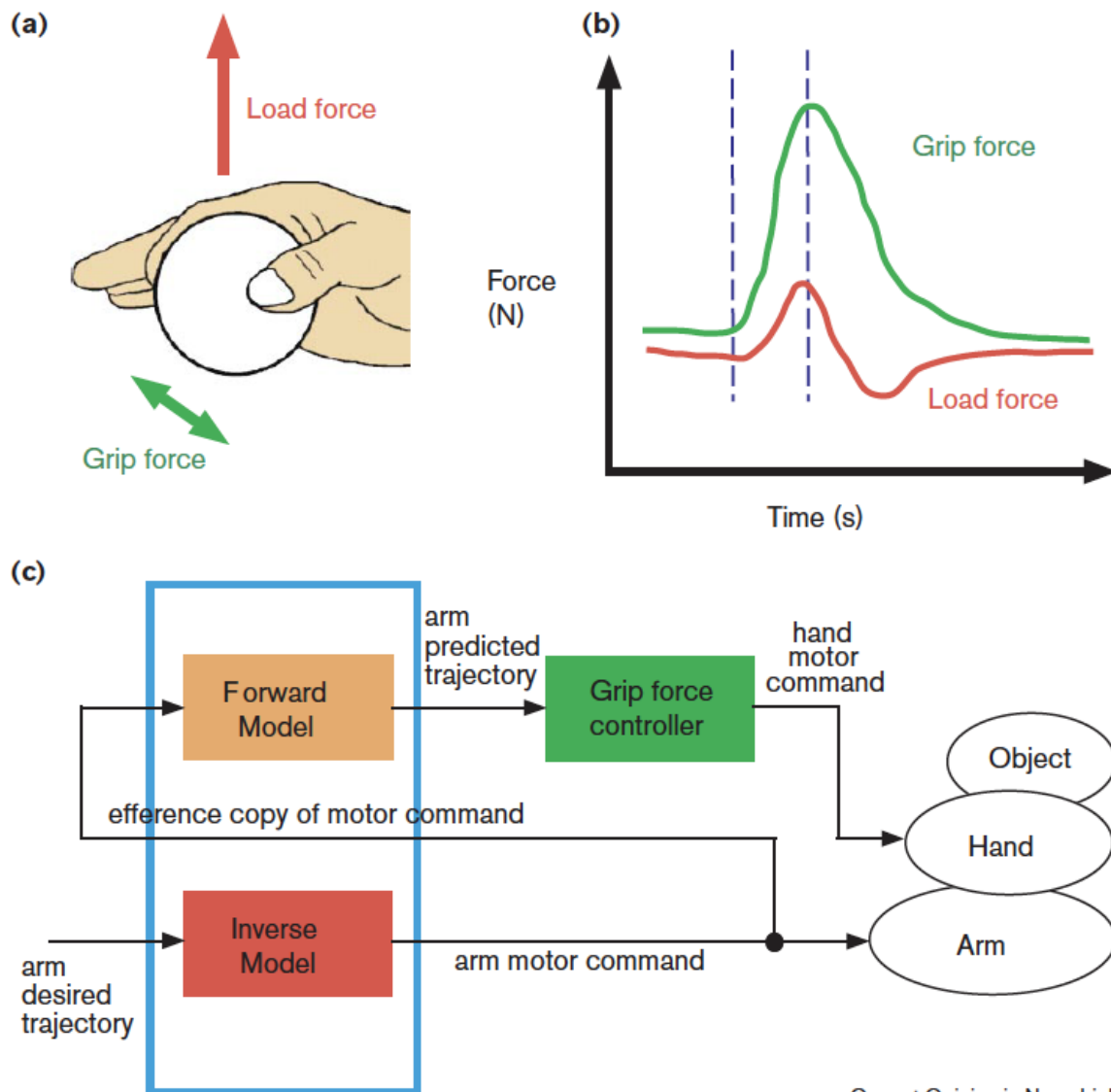
(Read Kawato paper)

(a)



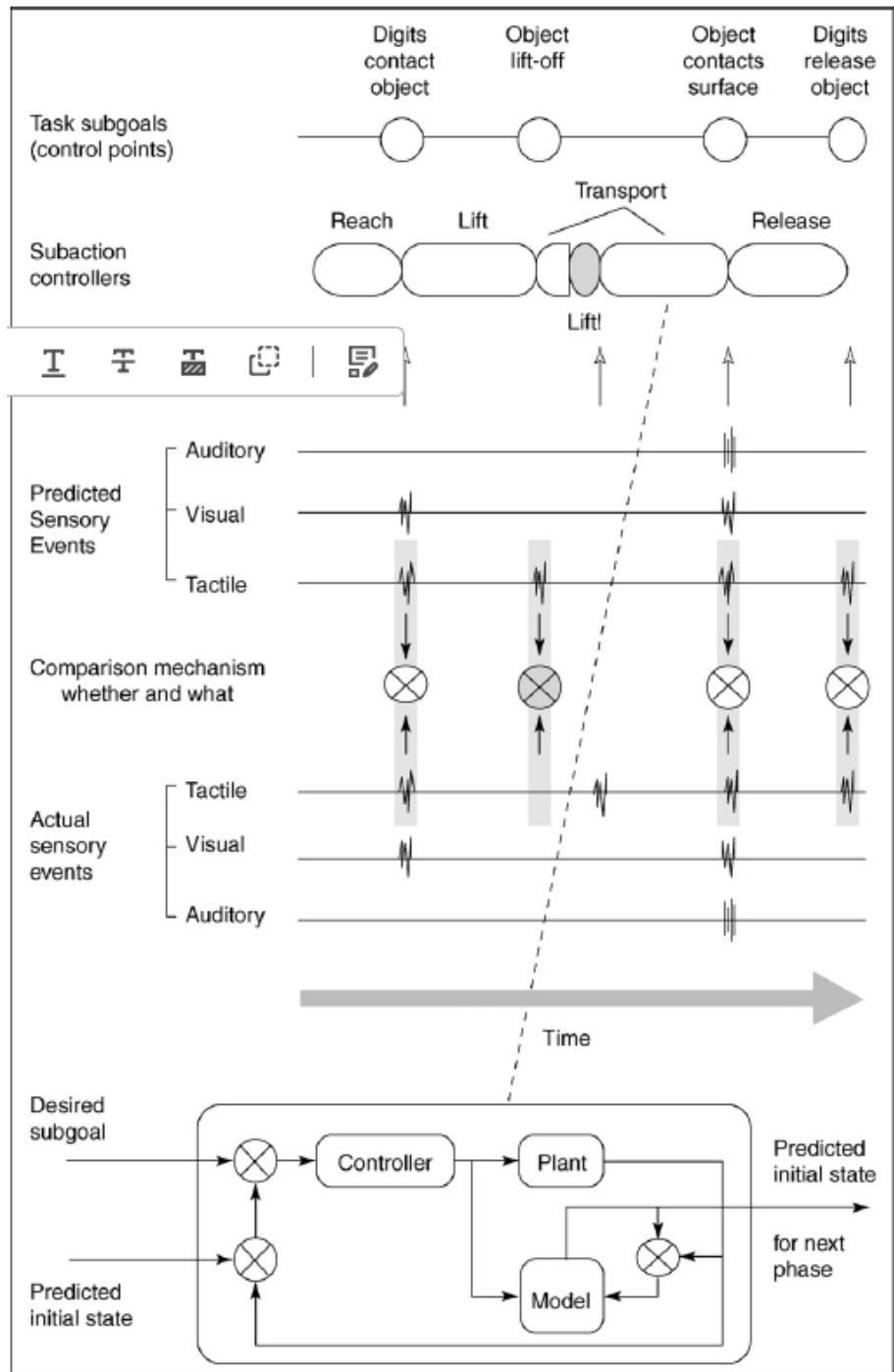
(b)





Current Opinion in Neurobiology

Coordination of grip force and load force, and a computational model based on internal forward and inverse models. (a) When an object is held with the tips of the index finger and thumb on either side, the normal grip force on the contact surface allows for the development of a frictional force, which prevents the object from slipping. (b) The grip force is precisely controlled so that it is just slightly greater than the minimum grip force needed to prevent slip. (c) This grip-force-load-force coupling is explained by a framework that contains both the inverse and forward models of the arm. Take, for example, a point-to-point arm-reaching movement that takes place while the hand is grasping an object. The inverse model of the combined dynamics of the arm, hand and object calculates the necessary motor commands from the desired trajectory of the arm. These commands are sent to the arm muscles as well as to the forward dynamics model as the efference copy. Then, the forward model can predict an arm trajectory that is slightly in the future. Given the predicted arm trajectory, the load force is calculated; then, by multiplying a friction coefficient and a safety factor, the necessary minimum level of grip force can be calculated. To realize this grip force time course, motor commands are sent to hand muscles. This model predicts that three different computational elements are necessary for grip-force-load-force coupling: the inverse model for arm reaching; the forward model for arm trajectory prediction; and the grip force controller for grip execution. Tamada *et al.* (T Tamada *et al.*, personal communication) have recently identified three fMRI active loci corresponding to these three elements.



Control Strategies in object manipulation tasks

Flanagan, Bowman and Johansson

Schematic illustration of the phases, separated by contact events, in a task in which an object is grasped, lifted off a surface, moved, and then replaced on the surface. The contact events shown at the top define subgoals of the task (i.e. goals of each action phase) and are signaled by distinct and discrete sensory events in one or more modalities. For example, when the object is replaced on the surface, the contact between the object and the surface gives rise to tactile, visual and auditory sensory events. By comparing these actual sensory events to predicted sensory events, the motor system can monitor task progression. For example, if the object is heavier than expected, the predicted tactile events associated with lift-off will not occur. As depicted in the figure, this triggers a second lift phase. (Although the figure only shows a comparison mechanism for tactile events, comparisons can be made for each task-relevant sensory event.) The predicted sensory events from each phase can be used to provide initial state estimates for the next phase. Predictive processes also operate within each phase. For example, during object transport, the motor system might use an internal model of the object in conjunction with a copy of the arm motor commands to predict the load force acting on the object, so that grip force can be suitably adjusted.